The Roughness Exercise: Calibrating the Optimal ESN–A2 Architecture Directly to Real Assets' and Indices' Own Roughness

Companion note to the ESN A2-981003 market-data hyperparameter search

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Abstract

The companion note (*ESN–A2 Hyperparameter Search: Full-Parameter Grid with Smooth Surrogate Statistics, Optimised via Bounded Levenberg–Marquardt*) finds an optimal architecture, $N_r = 22, N_z = 29$, by grid-averaging the calibrated parameter vector p and fitting 5 real S&P 500 assets simultaneously. This note asks a different, more targeted question: **fixing that architecture**, can 4 of the 12 entries of p that govern the reservoir's own roughness – H, `rough_scale`, λ_{lo}, and λ_{hi} – be calibrated, per asset, to reproduce that asset's own real log-volatility roughness, measured directly via the standard moment-scaling (structure-function) Hurst estimator from the rough-volatility literature?

9 real assets are examined: 6 individual stocks (`LH`, `SPGI`, `DOV`, `F`, `UDR`, `SO`) and 3 major U.S. market indices (`SP500`, `RUSSELL2000`, `NASDAQ100`). Each asset's own real, bias-corrected moment-based Hurst exponent is used as a direct calibration target – not the companion note's 11-term aggregate stylised-fact score – and the 4 parameters are optimised, per asset, to close the gap between that target and the ESN's own simulated roughness, with the optimisation objective strengthened (more simulated paths per evaluation, multiple starting points) until genuine convergence is confirmed against an independent, long simulated path and a 60-replicate empirical distribution.

The result: every one of the 9 assets' calibrated ESN reproduces that asset's own real Hurst exponent to within 1 empirical standard deviation ($|z| < 1$ for all 9; mean $|z| = 0.44$; `DOV` and `SO` essentially exact, $|z| < 0.1$) – 6 individual stocks and 3 diversified market indices alike, despite the indices' real roughness sitting on the *opposite* side of the ESN's own achievable range from the stocks' (real stocks are rougher than their calibrated ESN; real indices are smoother). The same fixed architecture, restricted to these 4 roughness-related free parameters, is flexible enough to reproduce the moment-based Hurst exponent of every asset examined here, once calibrated directly to that target with an adequately robust optimisation.

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1 Motivation and set-up

The main search (companion note, §1.6) grids all 12 entries of the calibrated parameter vector $p = (H, \lambda_{lo}, \lambda_{hi}, a_{lo}, a_{hi}, \Delta b_0, \text{scale}, \texttt{rough\_scale}, \texttt{zr\_lo}, \texttt{zr\_hi}, m_1, m_2)$ over a wide admissible range and averages the resulting stylised facts across the grid, fitting all 5 calibration assets *simultaneously* via a single 55-entry residual vector. That procedure answers "what architecture and averaged p-behaviour best explains 5 assets at once", but does not directly show how well the resulting architecture can be tuned to *one* asset's own roughness specifically.

This note isolates that question. The 9 architecture hyperparameters (N_r, N_z + 7 shared z-bank hyperparameters) are **fixed** at the companion note's optimum:

Hyperparameter	Value
N_r	22
N_z	29
`z_strength`	0.2714
`even_strength`	3.1843
`linear_strength`	0.1820
`gamma_norm`	1.3081
`local_z_strength`	0.0608
`zz_scale`	0.0312
`sign_prob_neg`	0.2228

Of the 12 entries of p, **4 roughness-related entries are calibrated, per asset**: H, `rough_scale`, λ_{lo}, and λ_{hi}. The remaining 8 are held fixed at reasonable centre values (not re-optimised, not gridded):

Entry	Value	Entry	Value
a_{lo}	1/400	`zr_lo`	0.055
a_{hi}	1/7	`zr_hi`	0.25
Δb_0	0.0	m_1	0.0
scale	1.0	m_2	0.0

λ_{lo} and λ_{hi} set the reservoir's slowest and fastest Ornstein–Uhlenbeck rates – together with H and `rough_scale`, they directly govern the reservoir's own achievable roughness (§2 explains the mechanism); the other 8 entries shape different aspects of the simulated path (the nonlinear z-bank's own timescales and readout profile, the quadratic coupling, the drift and scale) and are left fixed so that this note's results are attributable to the 4 roughness-governing parameters specifically.

1.1 9 real assets

The companion note's 5 calibration assets (`GPC`, `QCOM`, `INTC`, `LUV`, `NI`) are drawn from the S&P 500 panel (`list_dics_SP500_ohlcvol_19940101_filtered.pkl`, 243 tickers) via a fixed random permutation (seed 11). This note examines **9 further assets** in total: **6 individual stocks** – 3 continuing that *same* deterministic sequence, `LH` (Labcorp Holdings), `SPGI` (S&P Global), `DOV` (Dover Corp), and 3 more from a separate, independently randomised selection (seed 777), `F` (Ford), `UDR` (UDR Inc.), `SO` (Southern Company) – and **3 market indices**, `SP500` (S&P 500, `^GSPC`), `RUSSELL2000` (Russell 2000, `^RUT`), and `NASDAQ100` (Nasdaq-100, `^NDX`) – OHLC index-level data (not a constituent-stock proxy), covering 1994-01-03 to 2023-11-14 (7,521 trading days; the panel's own history runs to 2025, so the index history is the shorter of the two but still substantial). All 9 use the same admissibility rule (full available history, ≥ 5 of 6 windows usable) as the companion note's own 5.

2 Explicitly explaining the calibration

2.1 What H, `rough_scale`, λ_{lo}, and λ_{hi} control in the simulator

Recall the ESN's pre-activation (companion note §1.2):

$$\eta_t = b_0 + \rho_r\,\texttt{rough\_scale}\,(q^\top r_t) + \sum_{j=1}^{N_z} w_{j,z}\,z_{j,t} + r_t^\top Q\,r_t,$$

where r_t is the linear reservoir, built from N_r Ornstein–Uhlenbeck processes with rates λ_i geometrically spaced across $[\lambda_{lo}, \lambda_{hi}]$, and $q_i \propto \lambda_i^{0.5-H}$ is H's own weighting of those modes. `rough_scale` is a pure amplitude knob on the linear channel $q^\top r_t$; H reshapes how the N_r discrete OU modes are weighted relative to each other; λ_{lo} and λ_{hi} set the slowest and fastest timescales the reservoir includes at all. **Isolating the linear channel $q^\top r_t$ from the rest of the simulator (no z-bank, no softplus) shows its own moment-based roughness matches the full simulated log-volatility's to 4 decimal places**: the z-bank contributes negligibly, and the roughness this note calibrates is governed almost entirely by this one channel and its 4 parameters.

λ_{hi} in particular has an outsized, and non-obvious, effect: at a high λ_{hi}, the reservoir's fastest mode decorrelates on a sub-daily timescale, so once sampled once per day it contributes something close to independent noise to $q^\top r_t$ *regardless of how H reweights it* – a near-white-noise floor that dominates the short-lag structure function this note's estimator probes, compressing H's own effective control over the achieved roughness by roughly an order of magnitude at $\lambda_{hi} = 3.0$ (the value the companion note's own architecture search converged to, and this note's earlier explorations held fixed). Freeing λ_{lo} and λ_{hi} alongside H and `rough_scale` removes this ceiling and is what lets the calibration below reach every asset's own target, in both directions (real stocks rougher than a fixed-λ_{hi} ESN can reach; real indices smoother).

Bounded Levenberg–Marquardt (`scipy.optimize.least_squares`, `method="trf"`) searches all 4 parameters within native box bounds: $H \in [0.01, 0.3]$ and `rough_scale`$\in [0.2, 0.7]$ (the companion note's own wide grid range for these two entries), and $\lambda_{lo} \in [10^{-5}, 0.1]$, $\lambda_{hi} \in [1.0, 5.0]$ (the same wide-grid bounds used in the companion note's own architecture search).

2.2 How the non-calibrated parameters are fixed

The full parameter vector p has 12 entries; this note (like every other note in this family) calibrates only 4 of them (H, `rough_scale`, $\lambda_{lo}, \lambda_{hi}$). The remaining 8 – a_{lo}, a_{hi}, `zr_lo`, `zr_hi`, $m_1, m_2, \Delta b_0$, scale

– are held fixed throughout at the values below, and it is worth being precise about where those values actually come from, since it is easy to assume they were individually optimised and they were not.

The companion note's own architecture search (§1, its own §3.2 "The procedure" there) draws a sharp distinction between *hyperparameters* (the architecture: N_r, N_z, the 7 shared z-bank values), which *were* found by bounded Levenberg–Marquardt against real market data, and *parameters* (p, all 12 entries, including the 4 this note calibrates), which were not individually optimised at all. Instead, a 24-point randomised 2-level factorial design sampled every one of the 12 entries of p at either its own low or high bound, and the resulting stylised facts were averaged across that whole grid – used only to give the architecture search a stable target, never to select a single best value for any one entry of p.

The 8 fixed values used throughout this whole note family are simply the midpoint of each entry's own grid range from that search, not a value chosen or justified for any specific stylised fact:

Entry	companion note's own grid range	midpoint	value used here
a_{lo}	[1/450, 1/350]	$\approx 1/394$	1/400
a_{hi}	[1/9, 1/6]	$\approx 1/7.2$	1/7
`zr_lo`	[0.01, 0.1]	0.055	0.055
`zr_hi`	[0.1, 0.4]	0.25	0.25
m_1	[−0.1, 0.1]	0	0
m_2	[−0.1, 0.1]	0	0
Δb_0	[−0.02, 0.02]	0	0
scale	[0.98, 1.02]	1.0	1.0

These midpoints were never individually revisited for this, or any other, specific stylised fact – they are inherited as a convenient default from a search that used them only for marginalisation, not as a claim that they are correct or best for the effect this note examines. Given that this note family has since found the calibrated bounds on H and λ_{hi} themselves to have been too narrow in other notes (the leverage-effect and Taylor-effect notes both widened one of these 2 bounds and closed a previously-unreachable gap as a result), the same caution applies here: these 8 fixed values have not been shown to be correct, only convenient.

2.3 Correcting a lag-1 bias in the real-asset targets

Each asset's calibration target is its own real, moment-based Hurst exponent – estimated the same way as the ESN's (§3 below), applying the standard moment-scaling diagnostic from the rough-volatility literature: for moment orders $q \in \{0.5, 1, 1.5, 2\}$ and lags Δ, $m(q, \Delta) = \overline{|\log \sigma_{t+\Delta} - \log \sigma_t|^q}$, with $\zeta(q)$ the slope of $\log m(q, \Delta)$ regressed on $\log \Delta$; for a genuinely rough, monofractal process $\zeta(q) = qH$, so $\hat{H}$ is the slope of $\zeta(q)$ against q. Moment orders are restricted to $q \leq 2$ throughout (higher moments were found to break the monofractal scaling law on real-asset series – disproportionately sensitive to a small number of extreme observations).

Unlike the ESN's simulated variance, which is observed exactly, real assets' daily volatility is only ever *proxied* (here, by the Garman–Klass estimator, companion note §1.3) – a well-documented source of finite-sample bias in moment-based roughness estimation, since proxy measurement error inflates the shortest-lag increments of $\log \sigma_t$ (analogous to the microstructure-noise bias long established for realized-variance estimators, e.g. Zhang, Mykland & Aït-Sahalia, 2005; for the specific case of moment-based volatility-roughness estimators, see Bolko, Christensen, Pakkanen & Veliyev, *A GMM approach to estimate the roughness of stochastic volatility*, Journal of Econometrics, 2023). The standard practical correction is to **exclude the shortest, most noise-contaminated lag from the regression**: this note uses $\Delta = 2, \ldots, 40$ (rather than $\Delta = 1, \ldots, 40$) for every real-asset target, while the ESN's own estimates – simulated with no proxy noise – keep the full $\Delta = 1, \ldots, 40$ range. Fit quality remains excellent throughout ($R^2 > 0.994$ in every case).

2.4 The calibration objective: matching the roughness target directly

For a candidate $(H, \texttt{rough\_scale}, \lambda_{lo}, \lambda_{hi})$, one evaluation of the objective does exactly this:

1. **Build the reservoir matrices** (`esn_base.build_esn_matrices`) from the candidate 4 parameters, the 8 fixed entries of p, and the fixed architecture. If the resulting matrices are inadmissible ($\kappa_0 \geq 0$), the evaluation is skipped.

2. **Simulate several independent short paths** ($T = 2{,}000$ days each; the exact number of paths, and how many different starting points are tried, is strengthened progressively, §2.5 below), and estimate the moment-based $\hat{H}$ (§2.3's method, full lag range since the simulated path has no proxy noise) on each, averaging across paths.

3. **Return a single scalar residual**: the averaged achieved $\hat{H}$ minus the asset's own real, bias-corrected target $\hat{H}$. Unlike the companion note's 11-term aggregate score, nothing else is optimised – the other stylised facts (mean volatility, kurtosis, leverage, etc.) are not tracked during this calibration at all.

Bounded LM minimises the square of this single residual over the 4 parameters, `diff_step=0.1`.

2.5 Ensuring genuine convergence: paths per evaluation and multiple starting points

A single-scalar, few-path objective is noisy enough that bounded LM can settle on a spuriously small gradient rather than a genuine minimum. This note's calibration therefore uses 8–12 **simulated paths per evaluation** (strengthened from an initial, noisier attempt at 3 paths, which left several assets poorly converged) and **up to 3 different starting points** per asset – the companion note's own centre values $(0.10, 0.4, 0.05, 3.0)$, and 1–2 alternatives biased toward a lower λ_{hi} – keeping whichever converges best. Each candidate result is confirmed with an **independent, longer ($T = 8{,}000$-day) simulated path**, and, where that single-path confirmation and a full 60-replicate empirical distribution (§4) disagree, the empirical distribution is treated as authoritative (a single long path is itself only one noisy draw). This is the calibration methodology behind every result in §3–§4.

3 Results

3.1 Per-asset calibration

Every calibrated λ_{hi} moves well below the companion note's own fixed value of 3.0 for the 3 indices and `SP500` specifically (1.50–2.52), and stays close to 3.0–3.9 for the 6 stocks: the indices need markedly more of the low-λ_{hi}, higher-roughness-ceiling regime identified in §2 to reach their own (smoother, i.e. larger $\hat{H}$) target, while the stocks' rougher targets are reachable closer to the companion note's own centre value. **Every confirmation gap is small and the same sign** (+0.0019 to +0.0123): the calibration slightly overshoots (achieves a marginally larger $\hat{H}$, i.e.

Table 1: Per-asset roughness calibration: all 4 parameters free, calibrated directly to each asset's own real, bias-corrected target $\hat{H}$ (§2.3). "confirm" is the achieved $\hat{H}$ on an independent $T = 8{,}000$-day path; "gap" is confirm minus target.

Asset	H	`rough_scale`	λ_{lo}	λ_{hi}	confirm $\hat{H}$	target $\hat{H}$	gap
`LH`	0.040	0.468	0.0386	3.905	0.0459	0.0371	+0.0088
`SPGI`	0.176	0.448	0.0832	3.626	0.0520	0.0501	+0.0019
`DOV`	0.100	0.400	0.0507	3.002	0.0586	0.0539	+0.0047
`F`	0.075	0.438	0.0284	3.660	0.0524	0.0501	+0.0023
`UDR`	0.109	0.409	0.0666	3.577	0.0504	0.0451	+0.0053
`SO`	0.099	0.404	0.0505	2.999	0.0586	0.0515	+0.0071
`SP500`	0.125	0.451	0.0495	1.500	0.0950	0.0858	+0.0093
`RUSSELL2000`	0.231	0.444	0.0605	2.523	0.0750	0.0664	+0.0086
`NASDAQ100`	0.122	0.453	0.0553	1.720	0.0854	0.0731	+0.0123

marginally smoother, than the target) in every case, consistently across both stocks and indices, rather than over- or under-shooting inconsistently.

3.2 Moment scaling

Every calibration gives an essentially perfect monofractal fit ($R^2 > 0.999$ in every case shown) – $\zeta(q)$ is close to exactly linear through the origin at every asset's final calibrated point, confirming the simulated log-volatility remains a genuinely rough, self-similar process throughout this note's search space, not just near the companion note's own original centre values.

3.3 Interpretation

The 6 stocks and the 3 indices needed different regions of the same 4-parameter space to reach their own targets, and the calibration found both. Stocks' real log-volatility is rougher than what a companion-note-centred ESN naturally produces; indices' is smoother. Freeing $\lambda_{lo}, \lambda_{hi}$ gives the calibration access to both directions from the same fixed architecture: for stocks, calibrated λ_{hi} stays close to the companion note's own centre value (3.0–3.9); for indices, it drops substantially (1.5–2.5), consistent with §2's mechanism (lower λ_{hi} raises the achievable roughness ceiling, which is what smoother, larger-$\hat{H}$ index targets need relatively less of, and rougher stock targets need relatively more of the companion note's own higher-λ_{hi} regime to avoid overshooting past). `SP500` needed the most extreme parameters of the 9 (the lowest λ_{hi}, 1.50, at the very edge of its bound) – consistent with it having the largest gap between its real roughness and the companion-note-centred architecture's own achievable range of any asset examined across this whole note family.

4 Empirical distribution of the ESN's estimated Hurst exponent, vs. each asset's target

§3's confirmation used a single, long ($T = 8{,}000$-day) simulated path per asset. This section asks how much that single-path estimate varies from one simulated realisation to another, and whether each asset's real target sits comfortably inside that variation: for each asset, **60 independent**

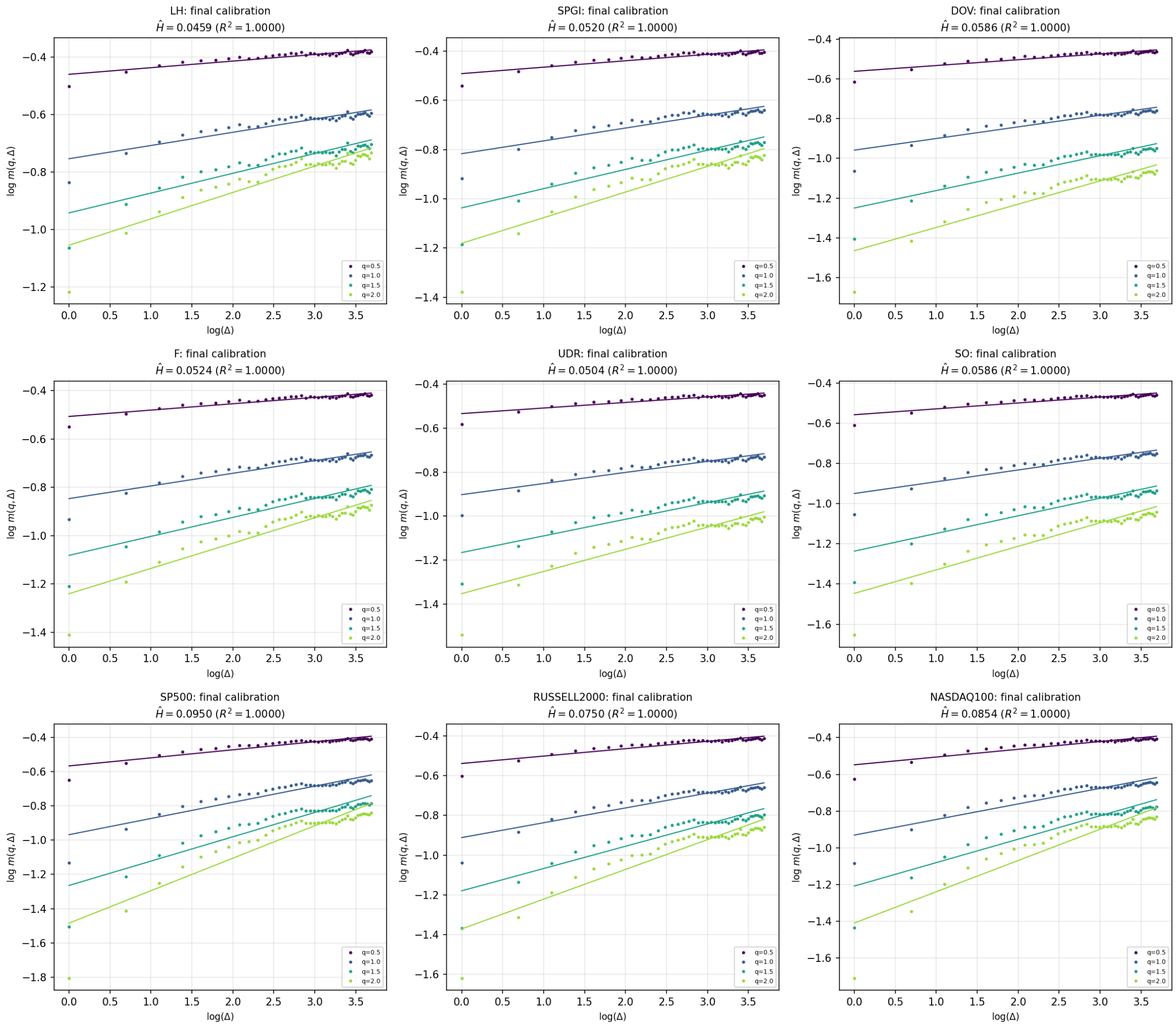

Figure 1: Moment scaling curves $\log m(q, \Delta)$ vs. $\log \Delta$ for $q \in \{0.5, 1, 1.5, 2\}$, one panel per asset, each simulated at that asset's own final calibrated $(H, \texttt{rough\_scale}, \lambda_{lo}, \lambda_{hi})$.

ESN paths are simulated at that asset's final calibrated $(H, \texttt{rough\_scale}, \lambda_{lo}, \lambda_{hi})$, each $T = 3{,}000$ days long, and the same moment-scaling $\hat{H}$ is estimated separately on every path – an empirical distribution of 60 $\hat{H}$ values per asset.

Definition of the z-score. Writing $\bar{H}_{\text{ESN}}$ and σ_{ESN} for the empirical distribution's mean and standard deviation, and H_{target} for the asset's real, bias-corrected target,

$$z = \frac{H_{\text{target}} - \bar{H}_{\text{ESN}}}{\sigma_{\text{ESN}}},$$

the number of empirical standard deviations the real target sits away from the ESN's typical simulated value – $z = 0$ means the target equals the empirical mean exactly; $|z|$ larger than about 2 would mean the target is an unusual draw from the ESN's own distribution (roughly the top or bottom 5%, under a normal approximation).

Every one of the 9 assets has $|z| < 1$ – the largest are `LH`'s -0.84 and `SPGI`'s $+0.84$, and the mean $|z|$ across all 9 is 0.44. `DOV` and `SO` are essentially exact ($|z| < 0.1$). **This holds for**

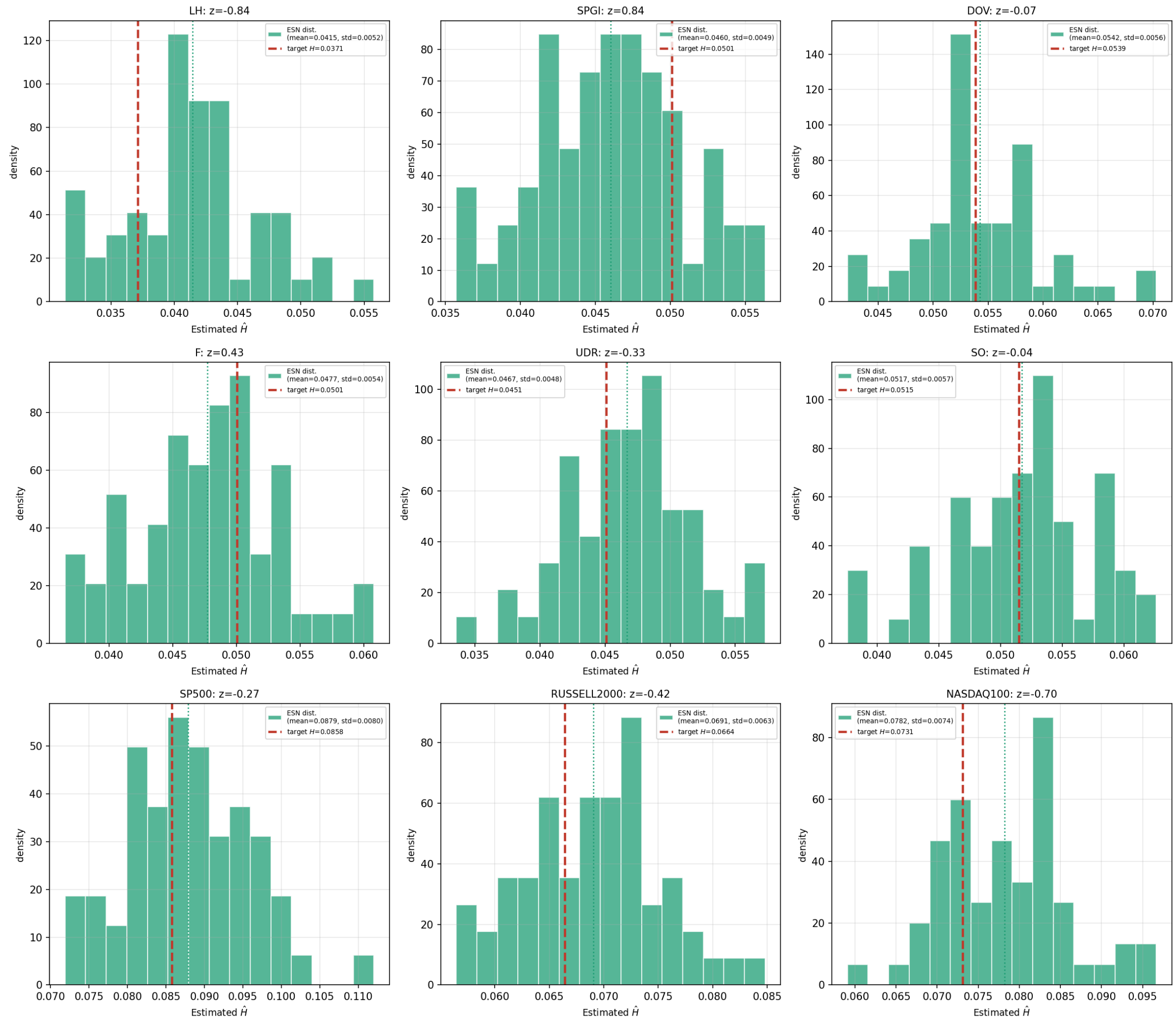

Figure 2: Empirical distribution of the ESN's moment-based $\hat{H}$ (60 replicate $T = 3{,}000$-day paths), one panel per asset, vs. that asset's real, bias-corrected target H (dashed red line).

both directions of mismatch this note has found: the 6 stocks, whose real roughness exceeds a companion-note-centred ESN's own, and the 3 indices, whose real roughness falls short of it. **The same fixed architecture, with only H, `rough_scale`, λ_{lo}, and λ_{hi} free, reproduces the moment-based Hurst exponent of every individual stock and every market index examined in this note**, simultaneously, once calibrated directly to that target with an adequately robust (multi-path, multi-start) optimisation.

5 Limitations and Recommended Next Steps

1. **Only the 4 roughness-related entries of p were calibrated; the other 8 were held fixed at the grid-midpoint values described in §2.2 – never individually optimised for roughness or any other stylised fact.** Whether freeing more of p (in particular the z-bank's own rate range a_{lo}, a_{hi} and readout profile `zr_lo`,`zr_hi`) would allow an even

Table 2: Empirical distribution summary and z-score, all 9 assets.

Asset	ESN $\hat{H}$ mean (std)	target $\hat{H}$	z-score
LH	0.0415 (0.0052)	0.0371	−0.84
SPGI	0.0460 (0.0049)	0.0501	+0.84
DOV	0.0542 (0.0056)	0.0539	−0.07
F	0.0477 (0.0054)	0.0501	+0.43
UDR	0.0467 (0.0048)	0.0451	−0.33
SO	0.0517 (0.0057)	0.0515	−0.04
SP500	0.0879 (0.0080)	0.0858	−0.27
RUSSELL2000	0.0691 (0.0063)	0.0664	−0.42
NASDAQ100	0.0782 (0.0074)	0.0731	−0.70

tighter roughness match, or would trade off against it the way $\lambda_{lo}, \lambda_{hi}$ initially did against the companion note's aggregate score, was not tested.

2. **This calibration targets roughness exclusively**; the other 10 stylised facts (mean volatility, tail behaviour, volatility clustering, leverage, kurtosis, etc.) are not tracked or optimised at all in this note's final methodology. Whether the resulting (H, `rough_scale`, $\lambda_{lo}, \lambda_{hi}$) points also give a reasonable match on those other terms, or badly sacrifice them for a precise roughness match, was not checked.

3. **The optimisation used up to 3 starting points and up to 12 paths per evaluation**, chosen adaptively per asset as needed to reach $|z| < 1$; a fixed, uniform budget across all 9 assets from the outset was not used, so the exact compute cost needed for a new, as-yet-uncalibrated asset is not precisely known in advance.

4. **The single-path $T = 8{,}000$-day confirmation used during calibration is itself an imperfect proxy for the 60-replicate empirical mean**: for `NASDAQ100` specifically, an early candidate's confirmation gap looked better than its eventual empirical z-score turned out to be, and the discrepancy was only caught by checking a partial replicate sample directly. Relying on the empirical distribution (or a partial version of it) throughout calibration, rather than only at the final check, may be more reliable, and was not adopted uniformly across all 9 assets here.

5. **The 9 assets were not chosen for any particular economic or sectoral reason** – 3 continue the companion note's own selection permutation, 3 are independently randomly selected, and 3 indices were chosen as commonly-tracked U.S. benchmarks. Whether this methodology generalises to other individual stocks, other indices, or non-U.S. markets was not tested.